SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50 Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **SWETA R (192511408)(**Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

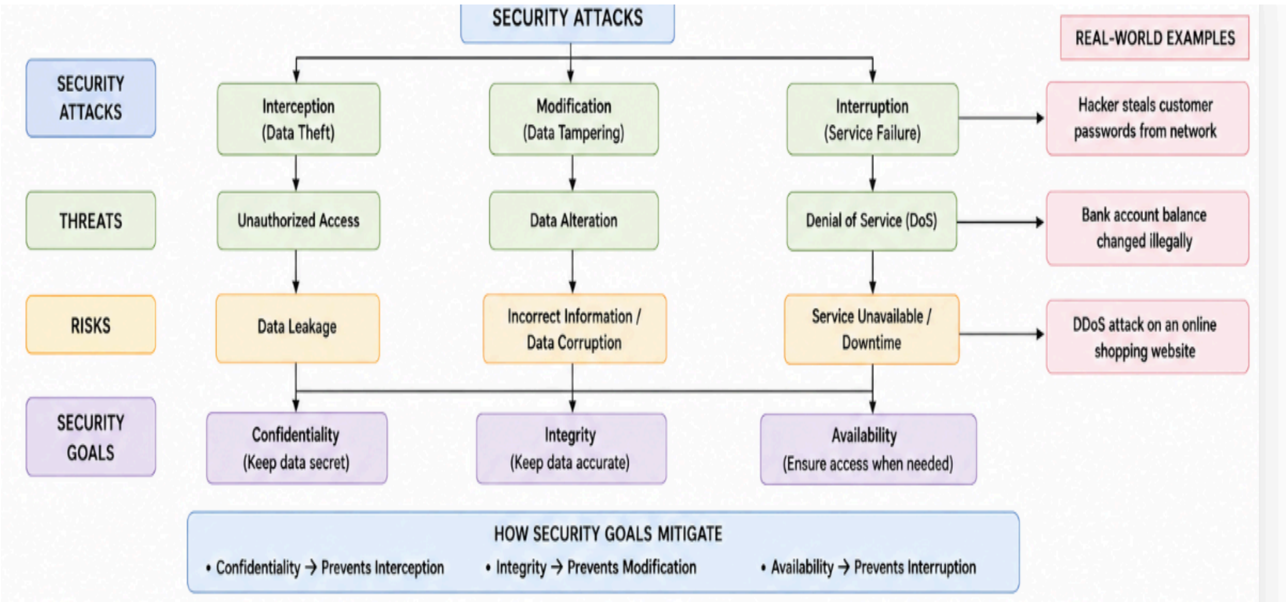
Signature of the student:Sweta R

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

a. Show how each attack leads to specific threats and risks
b. Map how security goals help mitigate them Provide one real-world Example



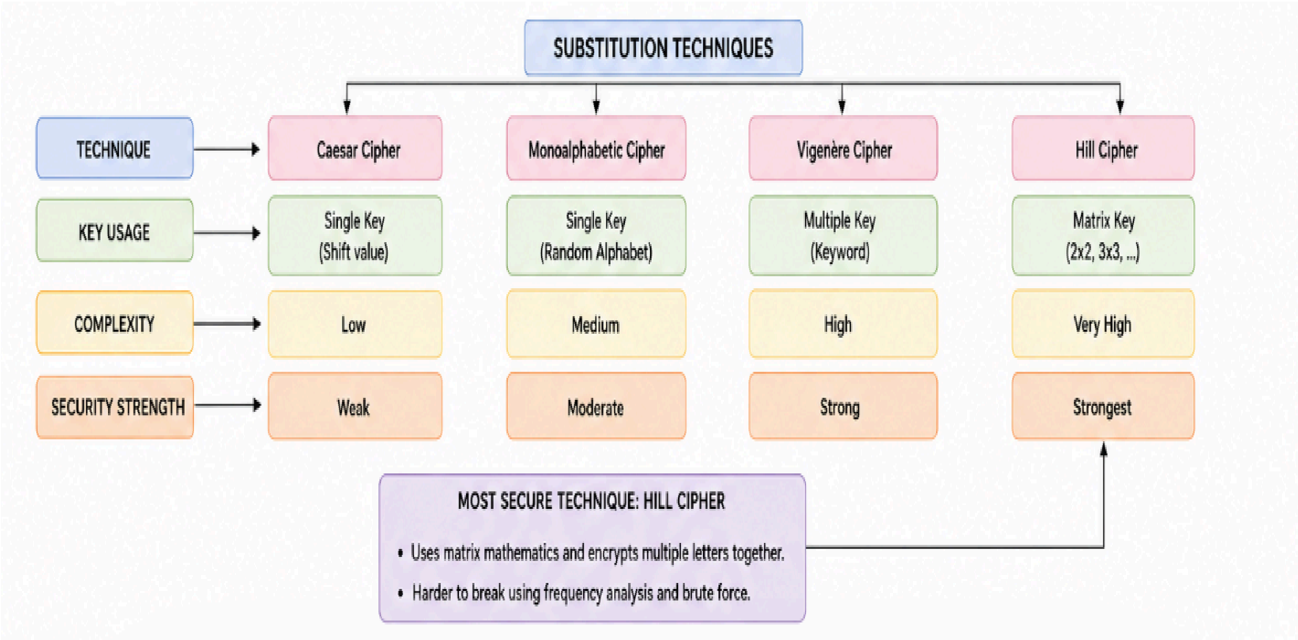
(a)

Attack	**Threat**	**Risk**	**Real-world Example**
Interception	Unauthorized access	Data leakage	A hacker steals login credentials from a database.
Modification	Data tampering	incorrect information	An attacker changes bank account balances.
Interruption	Denial of Service (DoS)	Service unavailable	A DDoS attack crashes an e-commerce

(b)

Security goal	**Mitigates**	**Real-world Example**
Confidentiality	Prevents unauthorized access	Encrypting patient medical records.
Integrity	Prevents data modification	Digital signatures protect online transactions.
Availability	Prevents service interruption	Backup servers keep banking services online

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



Analysis: Which technique is most secure and why?

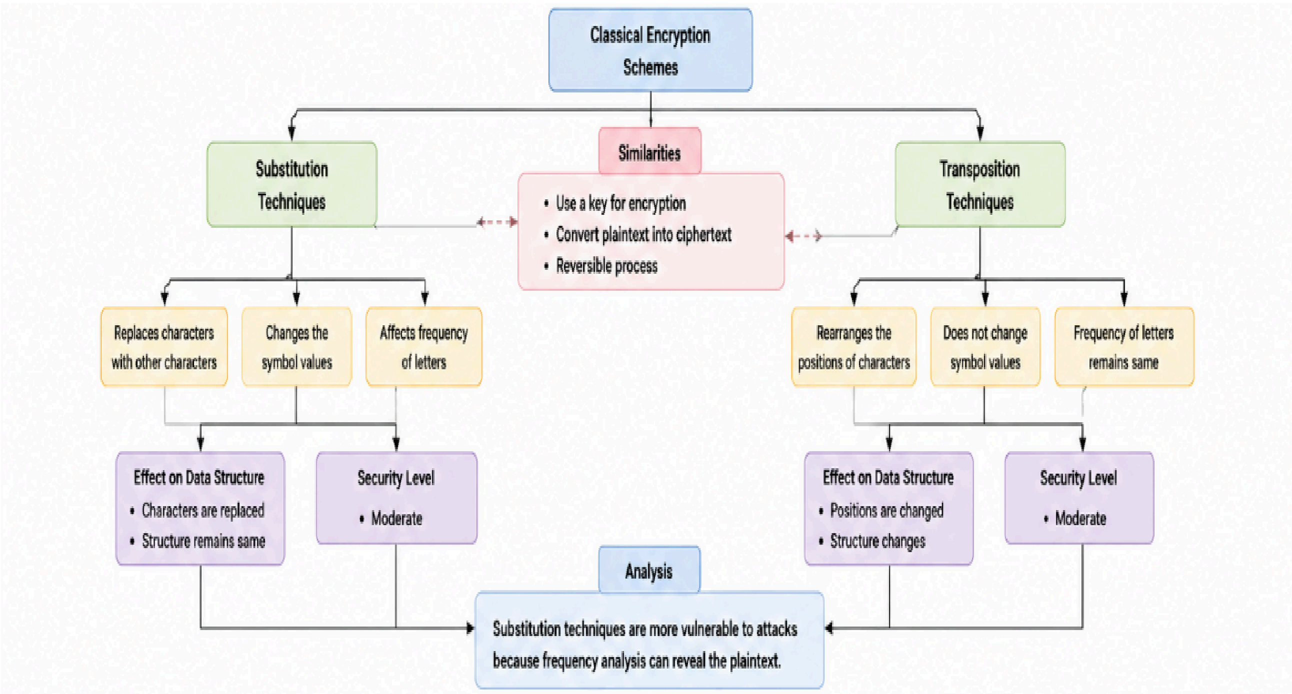
Hill Cipher is the most secure among these classical substitution techniques because:

- It uses a **matrix key** instead of simple letter substitution.
- It encrypts **multiple characters as a block**, making patterns difficult to identify.
- It is **more resistant to frequency analysis** than Caesar, Monoalphabetic, and Vigenère ciphers.
- Its mathematical operations make brute-force and cryptanalysis more difficult.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities

i.Map how each affects the Data structure and Security level

ii.Analyze which is more vulnerable to attacks and why



ii. Analysis: Which is More Vulnerable and Why?

Substitution techniques are generally more vulnerable to attacks because attackers can use **frequency analysis** to identify common letter patterns and recover the original message. Simple substitution ciphers, such as the Caesar cipher, are especially easy to break due to their limited key space. Transposition techniques are comparatively more resistant to frequency analysis since they preserve the original letter frequencies while only changing the positions of the characters.

Conclusion:

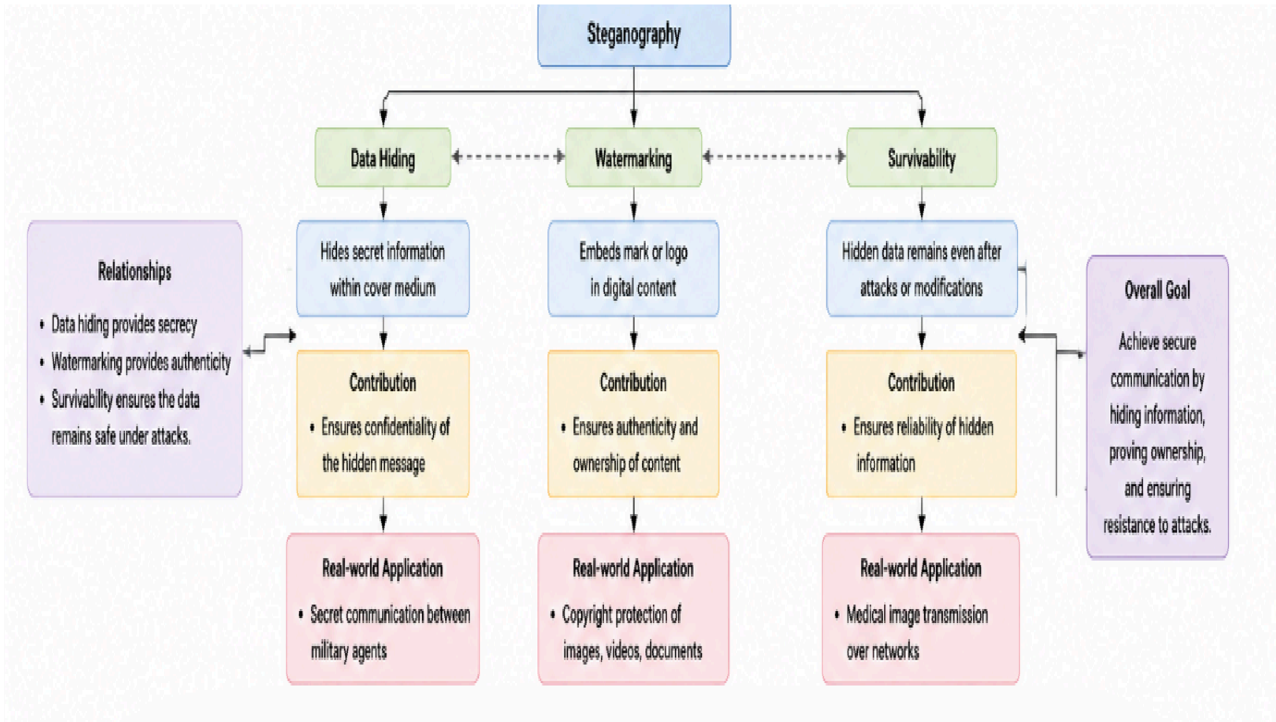
Both techniques improve data security, but substitution ciphers are generally more vulnerable than transposition ciphers. Combining substitution and transposition techniques provides stronger encryption than using either technique alone.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts

b. Analyze how each contributes to secure communication

c. Include one real-world application for each



a. Relationships Between the Concepts

The concept map shows that **Data Hiding, Watermarking, and Survivability** are the three major concepts of steganography. Data hiding conceals secret information, watermarking protects ownership and authenticity, and survivability ensures the hidden information remains intact after modifications or attacks.

b. Contribution to Secure Communication

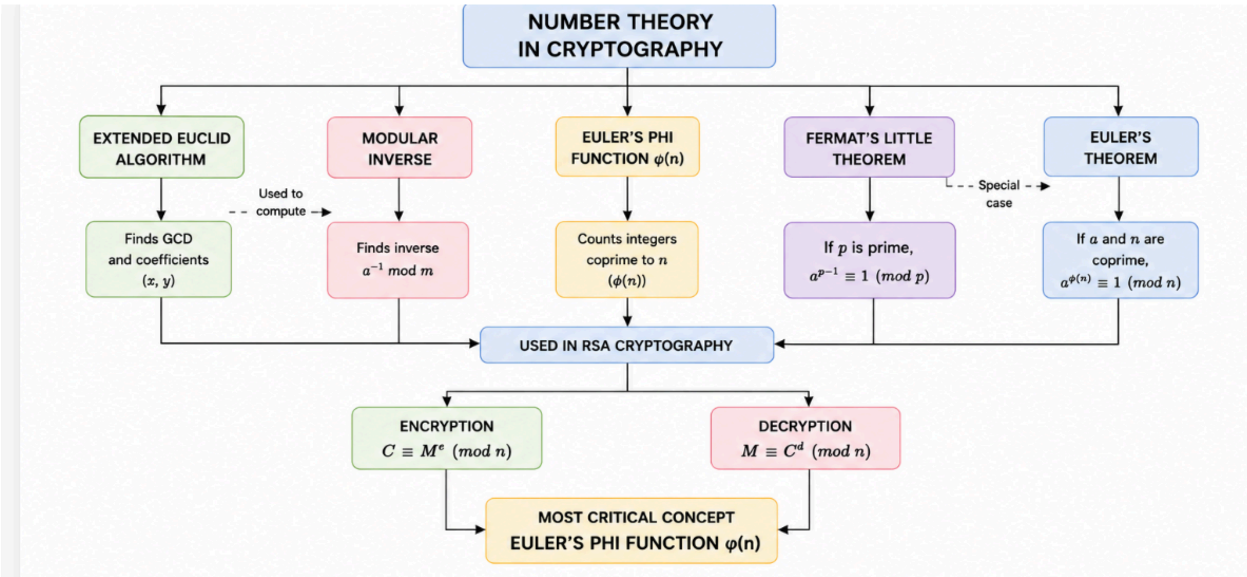
These concepts work together to provide secure communication. Data hiding maintains confidentiality, watermarking verifies authenticity and ownership, and survivability improves the reliability of hidden information during storage and transmission.

c. Real-World Applications

- **Data Hiding:** Secret military communication.
- **Watermarking:** Copyright protection for digital images and videos.
- **Survivability:** Secure transmission of medical images and confidential documents

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected
b. Map their role in encryption/decryption
c. Analyze which concept is most critical for public key cryptography and justify



a. Mathematical Connections

The concept map shows that the Extended Euclid Algorithm is used to find the Modular Inverse. Euler's Phi Function is used in Euler's Theorem, and Fermat's Little Theorem is a special case of Euler's Theorem.

b. Role in Encryption and Decryption

These concepts are widely used in RSA cryptography. Modular inverse helps generate the private key, while Euler's Phi Function and Euler's Theorem are used during encryption and decryption.

c. Most Critical Concept

Euler's Phi Function is the most critical concept because it is essential for RSA key generation and establishing the relationship between the public and private keys.

RUBRICS FOR CALCULATION

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understa ndi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/techno lo gies	Good understa ndi ng with proper integration of most concepts	Basic understan din g; limited or partial integration	Poor understan din g; unable to integrate concepts
Experim ent al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonab le interpreta tio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Documen tat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary	Basic documenta tio n with missing details	Poor or incomplete documenta tio n